

lab_code_12.c

```
1 // Name: Md Shagor
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4 // Lab Report: 12
5 // Experiment Name: Strong number calculation
6
7 #include<stdio.h>
8
9 int fact(int n)
10 {
11     int f = 1;
12     for(int i = n; i >= 1; i--)
13     {
14         f = f * i;
15     }
16     return f;
17 }
18
19 int strong(int dividend)
20 {
21     int temp, result, remainder, sum = 0;
22     temp = dividend;
23     while(dividend)
24     {
25         remainder = dividend % 10;
26         sum = sum + fact(remainder);
27         dividend = dividend / 10;
28     }
29     result = (temp == sum)? 1 : 0;
30     return result;
31 }
32 void range_strong()
33 {
34     int a, b, temp;
35     printf("Enter the minimum and maximum values of the range:");
36     scanf("%d %d", &a, &b);
37
38     for(int i = a; i <= b; i++)
39     {
40         temp = strong(i);
41         if(temp)
42         {
43             printf("%d \t", i);
44         }
45     }
46 }
47
48 int main()
```

```
49 {
50     int result, n, option;
51     printf("Enter 1 for single value check \n Enter 2 to find the strong numbers within a range.\n");
52     scanf("%d", &option);
53
54     switch(option)
55     {
56         case 1:
57             printf("Enter the number:");
58             scanf("%d", &n);
59             result = strong(n);
60             if(result)
61             {
62                 printf("%d is a strong number\n", n);
63             }
64             else
65             {
66                 printf("%d is not a strong number\n", n);
67             }
68             break;
69         case 2:
70             range_strong();
71             break;
72         default:
73             printf("Enter valid number");
74     }
75     return 0;
76 }
```